Lab Assignment: Mixed Diamond Pattern

Submission Details

- **Submission Platform:** Google Classroom
- **Submit:**
  - Your Java file (`DiamondPattern.java`)
  - At least 2 screenshots showing program execution

Objective

The goal of this assignment is to develop a Java program that generates an advanced diamond pattern. Students will practice the following concepts:

- Loop Structures (nested loops)
- Decision Structures (if/else)
- Logical Operators (`&&`, `||`)
- Modulus Operator (`%`)
- User Input using Scanner

Program Description

In this assignment, you will create a program that generates a **mixed diamond pattern** using both numbers and asterisks.

Program Requirements

User Input

- The program must ask the user to enter a number `n`.
- The number must be:
  - Positive
  - Odd
- If the input is invalid, the program must repeatedly ask for input using a **do-while loop**.
- You must use **logical operators** to validate input.

Pattern Rules

The program must print a symmetric diamond pattern based on the given number.

Example Output (n = 5)

```
  1
 ***
 12321
*******
123454321
*******
 12321
 ***
  1
```

Pattern Logic

- The pattern consists of two parts:
  - Upper half
  - Lower half
- Each row must contain appropriate **spaces** for alignment.
- If the row number is:
  - **Odd** → print numbers in palindrome form
  - **Even** → print asterisks (*)

Number Pattern Rule

For odd rows, numbers must be printed as:

- Increasing sequence from 1 to i
- Decreasing sequence from i-1 to 1

  Example:

```
12321
```

Asterisk Pattern Rule

For even rows, the number of asterisks must be calculated as:

```
2 * i - 1
```

Technical Requirements

- Use **do-while loop** for input validation
- Use **nested loops** for pattern generation
- Use **if/else** statements
- Use **logical operators**
- Use **modulus operator** to determine odd/even rows